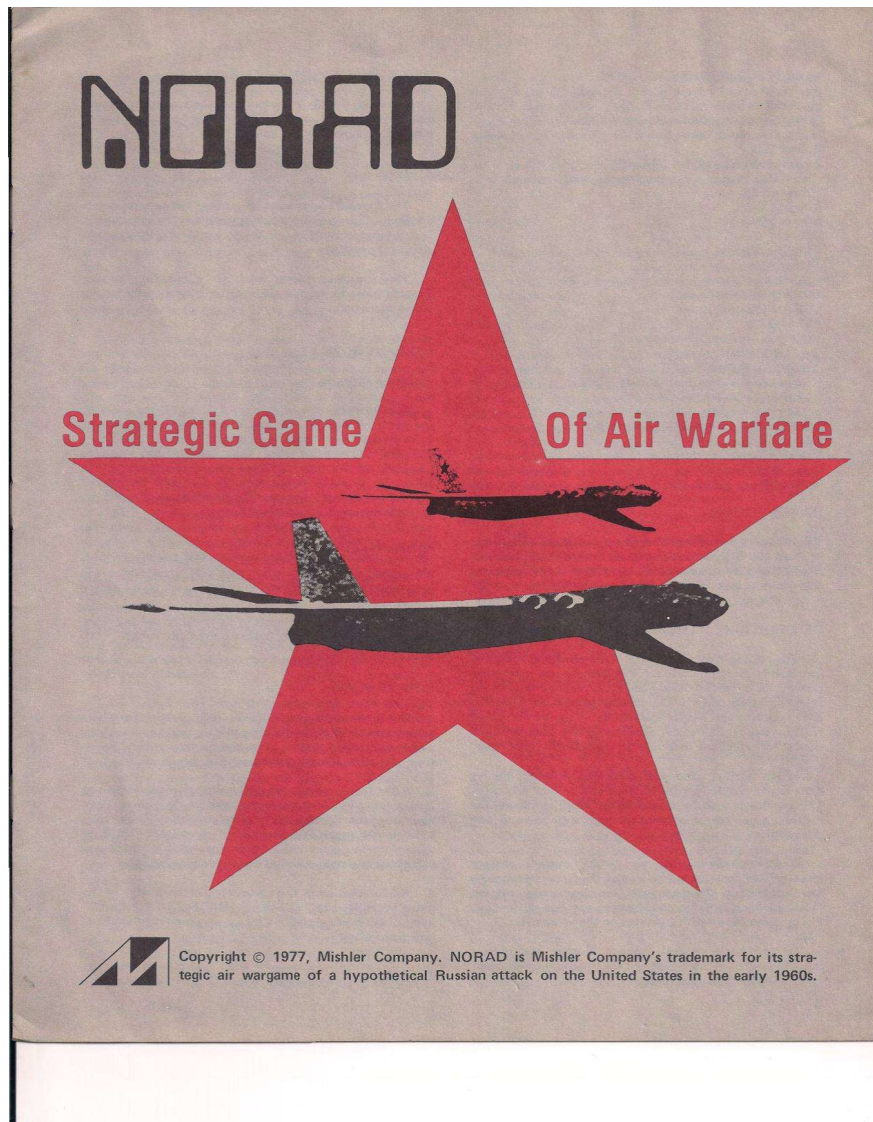




*Copyright © 1977, Mishler Company.
NORAD is Mishler Company's trademark for its strategic air wargame of a hypothetical Russian attack on
the United States in the early 1960s.*

Table of Contents

Introduction
Description of the Playing Pieces (Units)
Description of the Game Board
How to Win
Prepare to Play
American Player
Russian Player
How to Play
How to Move
Combat
Fighter Combat
Missile Combat
Bombing Cities
NORAD Solitaire Game
NORAD Optional Rules
The "DEW" Line
Optional Russian Placement
Part I: Siberian-Based
Part II: Cuban-Based
Russian Sub-Launched Ballistic Missiles
Canadian Air Defense
Assigned Targets
NORAD (1977) Revision Designer's Notes
NORAD Notes
NORAD Game Credits
Game Copy Number
Appendix: City Point Values
Appendix: Unit Reference

Introduction

NORAD is a representation of a hypothetical nuclear war between the United States and the Soviet Union in 1962. NORAD is a two-player game: one player is the American defender, the other player is the Russian attacker. [Note: Russian and Soviet are used interchangeably in these rules.

Description of the Playing Pieces (Units)

The playing pieces for NORAD are shown on the game board. Russian pieces are red and include real and decoy bomber units for the basic game, and sub-launched missile units for the optional game rules. American pieces are blue and include real and decoy fighter units, and real missile units. There are Canadian fighter units indicated with a "C" which are used only in the optional game rules.

Russian bomber units and American fighter units have a number printed on them next to their silhouette. This number is their movement factor, which represents the maximum number of squares the unit may move in any given Turn.

Russian pieces that are "real" units have a mushroom cloud printed on the opposite side of the piece. Decoy units have nothing printed on the other side. American units that are "real" have an explosion on the opposite side of the unit. Decoy units have nothing printed on the other side.

On the backs of real Russian bomber and missile units is a small number that is used to assign specific targets in the optional game rules.

Description of the Game Board

The game board covers most of North America. Russia and Cuba are off the map. The playing pieces are placed within the squares on the game board.

How to Win

The Russian player wins by gaining 100 or more points by bombing (destroying) American cities on the game board. The number of points each city is worth is printed in red next to that city.

The American player wins by eliminating enough Russian units to prevent the Russian player from gaining 100 or more points.

Prepare to Play

American Player

The American player sets up ALL of his/her units on the game board. The American units may only be placed on city squares. There are no restrictions on the number or types of American

units that may be stacked together on the city squares. The units are all placed face-up (fighter/missile silhouette showing).

Russian Player

The Russian units are staged face-up (silhouette showing) off the northern edge of the game board. The Russian player will be moving these units onto the map during his/her movement turns.

How to Play

The game is played in Turns with each player alternately moving, first the Russian player, then the American player (at his/her option) until one player wins. A player may only move his/her units during his/her Turn.

After the American player has finished placing all of his/her units, the Russian player may place as many Russian units as he/she wants on the map to begin the game. He/she must bring at least FOUR (4) units onto the game board each Turn until all thirty-one (31) of the Russian units have been brought into play.

All Russian units start from the northern edge of the game board (at the "RUSSIAN UNITS START HERE" line). These units may NEVER be stacked—only ONE (1) unit per square.

NOTE: *The furthest a Russian unit may move on its first move is to any square in row D. The minimum first movement of any Russian unit entering the game board is to any square in row B. The Russian player may bring his/her units onto the game map in a line—one in row B, one in row C, etc.*

All Russian units start and move face-up (bomber silhouette showing). They are only turned over (flipped) to show the nuclear mushroom cloud when they bomb an American city or are attacked by American fighters/missiles.

Once placed on the game board, no unit may leave the map unless it is eliminated in combat. Eliminated units, once removed from the game board, may NEVER be brought back.

NOTE: *If a Russian bomber unit reaches row V it stays in that square until the end of the game.*

After the Russian player moves all of his/her units (and any and all combat is resolved), the American player has the option of moving his/her fighter units, but is not required to do so. If the American player moves a fighter unit, it is removed from the game that same Turn. In other words, American fighter units are only used whenever a Russian unit comes within six squares of it.

How to Move

Every game board square requires one movement factor to enter. The Russian bombers and decoy units have a maximum movement factor of four. The American fighter units have a maximum movement factor of six. The "maximum movement factor" represents the maximum number of squares that a unit may move during each Turn. Note that units do NOT have to move their full movement.

All American fighter units may be moved in any direction or combination of directions (horizontal, diagonal, etc.) during the American player's Turn. American decoy fighter units and all missile units may NEVER be moved after initial placement.

All aircraft and aircraft decoys may move through squares occupied by enemy aircraft and aircraft decoys without having to stop or engage in combat. However, the Russian units must stop momentarily when they enter a square containing any American missile unit. When a Russian bomber moves on top of an American missile unit, the American player immediately announces whether or not the missile will fire. If it does not fire, the unit may bomb the city or continue on its way. Other Russian units may then move, even over the same city. As long as the missile unit remains on the game board, any new Russian units must also stop as soon as they enter the square.

Russian units may move in the same manner as the American fighter units except for the following restriction: At the end of the Russian player's Turn all bomber and decoy bomber units must be located at least TWO (2) rows further SOUTH than their original starting position for that Turn (unless the unit stops on a city square that Turn).

In other words the Russian units: (1) May NEVER move to the North-West, North and/or North-East; and (2) Must move at least TWO (2) squares to the South-West, South and/or South-East every Turn (unless they are going to bomb a city that Turn).

This applies to ALL Russian bomber and decoy units after they have been placed on the game board.

Combat

Units and cities on the game board are eliminated by combat. Combat may be fighter units attacking bomber units, missile units attacking bomber units, or bomber units attacking cities.

Fighter Combat

An American fighter unit may only attack a Russian unit by moving into the same square with it during the American player's Turn. This automatically eliminates the Russian and American units and both are removed from the game. This fighter combat is only resolved AFTER the American player has finished moving any of his/her fighters that are going to be moved for that Turn. In other words, American fighter units are only used whenever a Russian unit comes within six squares of it. Both the fighter and Russian units are turned over during combat (to prove that the fighter is not a decoy unit and to see if the Russian unit is a decoy or a real bomber) before they are BOTH removed from the game.

NOTE: *Fighter units may move through a square with a Russian bomber and not attack it. Also note that fighters may only attack bombers on the American movement Turn. Fighters on a city square may NOT attack Russian units which enter that square during the Russian player's movement Turn.*

Missile Combat

American missile units may eliminate any Russian unit that moves into the square they are in. The American player has the option to use the missile or save it. If the missile is used (attacks the Russian unit) the missile and the Russian units are turned over (to see if the Russian unit

was a decoy or a bomber) and then BOTH units are immediately removed from the game. If the missile unit is not used, the Russian player may continue moving his bomber unit or attack (bomb) the city.

NOTE: A Russian unit may NOT bomb a city if it was destroyed by a missile unit as it entered the city square. However, ANOTHER bomber unit may move to that same city during the same Turn, and if not attacked by any other missile units on that city square, it may destroy the city (flip the bomber over to show the mushroom cloud).

Bombing Cities

An American city may be destroyed whenever a Russian bomber unit (with a nuclear mushroom cloud on its reverse side) stops in the same square as that city and is not attacked by an American missile unit. At this time the Russian player may decide whether or not to bomb the city. If the city is going to be attacked the Russian unit is flipped over to show the nuclear cloud and that city's points are given to the Russian player. A Russian player may only destroy and receive points for a city ONCE. After a Russian unit bombs a city the unit may NEVER be moved again (and therefore never attack any other city).

NOTE: Keep the Russian nuclear mushroom cloud unit on the city square to help keep track of the total number of cities destroyed and gained by the Russian player.

If there are any American units on a city square when the square is bombed, those units are also eliminated and removed from the game. These eliminated units may not be checked by the Russian player to see if they are decoys until the game is over.

American decoy units may NEVER attack any Russian units. They are only removed from the game when the city they are on is destroyed. Russian decoy units may NEVER attack and are turned over and eliminated only when attacked by fighter or missile units.

If two or more fighter units attacking a Russian bomber are revealed to be real fighters, the extra fighter units are also eliminated (without being able to move or attack any other unit). Remember—fighter units are not turned over until ALL fighter units that you wish to move for that Turn have been moved.

Also note that unless the Russian units are moved in a logical manner, the Americans will always win.

NORAD Solitaire Game

NORAD can be played by just one person with computer AI controlling one side.

NORAD Optional Rules

The following optional rules may be added to make NORAD a more accurate simulation.

The “DEW” Line

With this optional rule the two northern-most cities Anchorage and Godthåb represent the key-stone radar stations in the “DEW” Line.

If the Russian player destroys BOTH of these cities, the radar integrity of the “DEW” Line is interrupted and radar coverage of the northern approaches is reduced to zero above the H row. The Russian may then stage his/her bombers on the game board at the H row instead of the North staging area as in the basic game. This means the Russian bombers could move from staging on row H, as far south as the L row, if Anchorage and Godthåb are destroyed.

Optional Russian Placement

Part I: Siberian-based

The Russian Siberian-based units may start at the western edge of the game board (at the “SIBERIAN-BASED OPTIONAL START LINE”). Any number of Russian units may start from this line, in addition to the “RUSSIAN UNITS START HERE” line.

Part II: Cuban-based

The Russian player may place a force of no more than five (5) units in Cuba, off the southern edge of the game board. No more than three (3) of these units may be real bombers (the remaining units must be decoys). These aircraft are actually the Il-28 “Beagles” based in Cuba. They must be placed on the game board in the “CUBAN-BASED PLACEMENT BOX” BEFORE the American player places his/her units on the map.

The Russian player may bring any number of these Cuban-based units into play during any of his/her Turns by placing them on the squares of the V row directly to the north of the “CUBAN-BASED PLACEMENT BOX.” They may move no farther than that square (on row V) on their first move. Units in the “CUBAN-BASED PLACEMENT BOX” may NOT be started from any other location except the squares touching the “CUBAN-BASED PLACEMENT BOX.”

On the Russian player’s next Turn, the Cuban-based units may move their normal movement (maximum of four squares per Turn). However, the Cuban-based Russian units have a reverse movement restriction from their comrades starting in Russia: Cuban-based units must end their movement each Turn at least TWO (2) rows further north than they started, unless they bomb a city that Turn.

Russian Sub-Launched Ballistic Missiles

With this rule the Russian player adds the three sub-launched missile units and one decoy sub-launched missile unit provided on the unit counter sheet. These units are shown and listed under the “OPTIONAL NORAD PLAYING PIECES” on the game board. These are the “Sark” SS-N-4 (formerly “Snark”) missiles carried by the Russian ballistic-missile Z-V class submarines. The Z-V subs have two vertical launchers in the conning tower and no more than six to ten were available in 1962. The “Sark” carries a one megaton nuclear warhead 372 miles.

The Russian player may bring any one or more of these units onto the game board during any of his/her Turns. He/she starts these units by placing them (no more than one unit per square) one (1) square from the coast of the United States; i.e., adjacent to Norfolk, adjacent to New York, adjacent to San Diego, etc. During the American player’s Turn these Russian missile units may be attacked by American fighters in the same manner as fighters attack bombers.

On the Russian player's next Turn, the missile units may move one square and destroy any American city they move onto. They must be removed from the game or attack a city the Turn after being placed on the game board. Remember: Russian missile units may move no more than one (1) square after being placed on the game board. The decoy sub-launched missile unit may also move one square to simulate an attack on a city. While in a city square the Russian missile units may be attacked by American missile units (in the same manner as bombers are attacked when they move into city squares that contain American missile units). Each Russian sub-launched missile unit may only attack one city.

Canadian Air Defense

With this rule, three real Canadian fighters and one decoy fighter unit are brought into the game. The Canadian units may only be placed on any Canadian (not U.S.) cities to start the game. When this optional rule is selected, American units may only be set up in U.S. cities.

Canadian fighter and decoy units are used in the same manner as the American fighter and decoy units by the American player. This optional rule must be included if the Russian player adds his/her sub-launched missile units to the game.

Assigned Targets

After the American player has placed all of his/her units on the game board the Russian player assigns a city to each one of his/her real bombers (numbered from one to twenty-three). Each bomber may only bomb the city that has been assigned to it.

If the optional Russian sub-launched missiles are used, then the Russian player must also assign cities to the three real ballistic missile units. This is done for the Russian missiles BEFORE the American units are placed on the game board. The city listed is the only city the Russian sub-launched missile unit may attack.

Appendix: City Point Values

Point values for American and Canadian cities shown on the game board, as transcribed from the map sheets. A dash indicates the value was not legible on the scanned map.

City	Point Value
Anchorage	5
Vancouver	5
Seattle	7
Portland	5
San Francisco	7
Los Angeles	6
San Diego	8
Phoenix	5
Denver	6
Edmonton	5
Winnipeg	5
Minneapolis	5
Chicago	9
Omaha	9
Kansas City	6
St. Louis	5
Dallas	5
San Antonio	5
Houston	5
New Orleans	6
Mobile	6
Birmingham	6
Key West	6
Atlanta	5
Jacksonville	8
Savannah	5

City	Point Value
Norfolk	7
Washington, D.C.	9
Philadelphia	5
Pittsburgh	7
Cincinnati	6
Toronto	6
New York	9
Boston	5
Quebec	5
Godthåb	5
Detroit	8

Appendix: Unit Reference

Standard NORAD Playing Pieces

Unit	Quantity	Movement Factor
American Fighter Unit	x12	6
American Fighter Decoy Unit	x4	—
American Missile Unit	x5	6
Russian Bomber Unit	x23	4
Russian Bomber Decoy Unit	x8	4

Optional NORAD Playing Pieces

Unit	Quantity
Canadian Fighter Unit	x3 real + 1 decoy
Russian Missile Unit	x3 real + 1 decoy